

Vaughn Lee Edwards Jr.

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SUMMARY

Computer Science and Data Analytics student at James Madison University building machine learning systems and analytics platforms that solve real business problems. Skilled in Python, SQL, XGBoost, and AWS infrastructure with a focus on production deployment and cost-quality tradeoffs. Driven by infrastructure optimization and infrastructure challenges, currently developing an LLM cost router that predicts when cheaper models suffice, extending professional LLM evaluation experience from Handshake AI Research Program. Approach technical work with rigor: validate assumptions, identify and fix systematic errors, prioritize business impact over feature count.

EDUCATION

- **James Madison University** — Rising Junior (Computer Science | Data Analytics) – 3.2 GPA | Est. May 2028
 - *Relevant Coursework:* Introduction to Data, Applied Statistics Using R, Algorithms & Data Structures, Discrete Structures, Calculus I

PROJECTS

- **NBA Player & Team Analytics Dashboard** | PostgreSQL, Excel, Power BI, R, PowerPoint | <https://github.com/vledwardsjr/NBA-Analytics-Project> | June 2026
 - Designed and normalized a PostgreSQL database spanning 6 NBA seasons of player and team statistics; built all scoring logic in SQL using composite performance metrics
 - Identified systematic formula errors in the ranking system and corrected them; reduced award categories from 9 to 5 after diagnosing small-sample bias
 - Delivered interactive Power BI dashboards, technical methodology report, and stakeholder presentation; published complete project to GitHub
- **LLM Cost/Latency Router** | Python, XGBoost, SQL, AWS | <https://github.com/vledwardsjr/LLM-Router-Project> | In Progress - Est. August 23, 2026
 - Built predictive routing model that determines whether queries need an expensive LLM or can be handled by a cost-efficient model, enabling companies to optimize inference cost vs. quality tradeoffs at scale
 - Engineered query-level features (complexity, reasoning steps, entity density) from 2,000–5,000 labeled queries scored by LLM-judge and manually validated; trained XGBoost vs. Logistic Regression with ROC-AUC and feature importance analysis
 - Deployed as serverless API endpoint with Pareto frontier analysis quantifying cost savings vs. quality loss; deliverables include final report, threshold tuning analysis, and GitHub documentation
- **NYC Jobs Market Analysis Dashboard** | Excel | <https://github.com/vledwardsjr/nyc-jobs-excel-dashboard> | February 2026
 - Built interactive dashboard using Excel (Power Query, PivotTables) on cleaned job market dataset spanning 1,000+ records
 - Conducted exploratory analysis across salary, hiring trends, and market variables to identify key patterns in job demand and compensation
 - Published dashboard, analysis, and visualizations to GitHub to communicate findings for data-driven hiring decisions

SKILLS

- **Programming & Data Tools:** Python (pandas, scikit-learn, XGBoost) | SQL (window functions, CTEs, layered view architecture, relational database design) | Excel (PivotTables, Power Query, advanced formulas, data cleaning) | Tableau (dashboards, data visualization) | Power BI (dashboard development, DAX) | R (R Markdown, data analysis, technical reporting) | Streamlit (interactive ML demos)
- **Data Analytics & Methods:** Data Cleaning & Transformation (ETL concepts) | Exploratory Data Analysis (EDA) | Machine Learning (classification, feature importance, error analysis, predictive modeling) | NLP & Text Feature Engineering | Model Evaluation (precision, recall, ROC-AUC, SHAP) | Data Visualization & Dashboarding | KPI Development & Performance Metrics | Statistical Analysis (regression, distributions) | Data Modeling (Dimensional/Relational Modeling) | Cost-Quality Tradeoff Analysis
- **Other Tools:** Git & GitHub (version control) | Jupyter Notebook | Relational Database Design | AWS (Lambda, API Gateway, DynamoDB) | API Development | Serverless Computing
- **Core Competencies:** Analytical Thinking & Problem Solving | Data Interpretation & Communication | Attention to Detail | Cross-Functional Collaboration

CERTIFICATIONS

- IBM Intro to Data Analytics (2025) | Excel for Data Analytics (2025) | SQL for Data Analytics (2025) | SQL+ for Data Analytics (2026) | Power BI for Data Analytics (2026) | CompTIA IT Fundamentals (ITF+) | IC3 GS6 Level 1 & Level 2
- *Global Career Accelerator* — Data Analysis (Excel) (Mar 2026) | Data Visualization (Tableau) (May 2026) | AI Professional Skills (May 2026) | Intercultural Skills (May 2026)

WORK/LEADERSHIP EXPERIENCE

- **Fit Right Kid's Shoes Inventory Specialist (September 2022 - August 2024)**
 - Managed and tracked inventory data across 100+ items using POS systems
 - Ensured data accuracy through validation and consistent data entry processes
 - Supported inventory analytics by organizing and updating product records
- **Fit Right Kid's Shoes Front Desk & Customer Data Specialist (November 2024 - January 2026)**
 - Maintained and validated customer data using POS and scheduling systems
 - Ensured data accuracy and consistency for operational efficiency
 - Collaborated with team members to improve data reliability in scheduling workflows
- **Bridgely (Early-Stage Startup) — Co-Founder & Operations Lead**
 - Led data validation and organization of operational datasets to ensure accuracy and usability
 - Coordinated workflows and managed data processes to support platform functionality
 - Contributed to early-stage data structuring and process optimization
- **Handshake AI Research Program — AI Trainer (February 2025 - May 2026)**
 - Evaluated large language model outputs using structured evaluation frameworks to assess accuracy and reasoning quality
 - Analyzed response patterns across multiple prompts to identify inconsistencies and improve model performance
 - Applied analytical thinking and attention to detail in high-volume evaluation workflows

INTERESTS

NBA & Sports Analytics | Fitness & Weightlifting | Exploring Data Projects and Visualization | AI/ML technology